

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is **5** square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

Answer

- A. **7**
- B. **10**
- C. **20**
- D. **25**

Correct Answer: C

Rationale

Choice C is correct. It's given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It's given that the length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is **2**. It follows that the scale factor between their areas is **2²**, or **4**. It's given that the area, in square centimeters, of parallelogram $ABCD$ is **5**. It follows that the area, in square centimeters, of parallelogram $PQRS$ is **5(4)**, or **20**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.